

APPROVED METHOD:
FOR PHOTOMETRIC TESTING OF
ENTERTAINMENT LIGHTING LUMINAIRES
USING INCANDESCENT FILAMENT LAMPS OR
HIGH INTENSITY DISCHARGE LAMPS
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-73-18(R2023)

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FILAMENT LAMPS OR HIGH
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AN AMERICAN NATIONAL STANDARD**

Publication of this Committee Report
has been approved by the IES.
Suggestions for revisions should
be directed to the IES.

**Prepared by
of the IES Testing Procedures Committee**



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Foreword

This Approved Method describes a standard procedure by which entertainment lighting luminaires, specifically designed for use in the theater, TV environment, film studios, or on-location shoots, can be measured.

Entertainment lighting luminaires usually have much narrower beam spreads than other luminaires, such as those covered by IES LM-35, *Approved Method for Photometric Testing of Floodlights using HID or Incandescent Filament Lamps*.¹ Therefore, different test methods and forms of presentation may be used. For example, test data is required in the form of illuminance values on a vertical aiming plane. Data reporting is covered by publication ANSI E1.9-2007 (R2017), *Reporting Photometric Performance Data for Luminaires Used in Entertainment Lighting*, developed by the Entertainment Services and Technology Association.² IES LM-73 is to be used in conjunction with ANSI E1.9.

In 1998, the IES assembled a task force composed of light measurement experts and members of the theater and TV community. This task force was charged with writing an Approved Method for testing entertainment lighting luminaires, resulting in this document.

It should be noted that laser systems and scenic projectors are not covered by this Approved Method.

1.0 Selecting Luminaires

Luminaires selected for testing shall be representative of the manufacturer's typical product, unless the specific purpose of the test is to evaluate other conditions.

2.0 Basic Laboratory Conditions

2.1 Stray Light

Precautions shall be taken to eliminate stray light from the test environment by the use of adequate shielding

and baffling. For moving cell photometers, the only light incident on the detector shall be that which is directly emitted from the luminaire. For a moving mirror photometer, only light from the luminaire reflected via the mirror to the detector shall be measured.

2.2 Ambient Temperatures

The ambient temperature of the photometric laboratory shall be maintained at $25^{\circ}\text{C} \pm 5^{\circ}\text{C}$ ($77^{\circ}\text{F} \pm 9^{\circ}\text{F}$).

2.3 Test Lamp Selection and Seasoning

Test lamps shall be selected for conformance to the manufacturer's design dimensions and construction. The lamps shall be seasoned in accordance with LM-54, *IES Guide to Lamp Seasoning*,³ so that their characteristics will be nearly constant during the test.

2.4 Test Lamp Operation

Stable lamp operation is necessary. Incandescent filament lamps shall be operated at a constant current corresponding to the current required to produce approximately 75 percent of rated lumen output. A stabilization time of 10 minutes is usually sufficient for incandescent lamps. Discharge lamps shall be operated at their rated wattage and shall be run until their light output is stable.⁴ Identical operating conditions must be used for the luminaire test and for lamp calibration.

If absolute measurements are required, incandescent lamps (including tungsten-halogen), shall be operated at full rated voltage, and HID lamps shall be operated at full rated wattage.

As an alternative method, for situations where lamp wattage cannot be conveniently measured, discharge lamps may be operated on the manufacturer's specified ballast and at rated voltage for both the luminaire test and lamp calibration. This technique must be noted on the test report, and may produce results slightly different from those obtained when stabilization is based on lamp wattage.

2.5 Test Distance

The distance between the luminaire and the detector shall be sufficiently large so that the inverse-square law (ISL) can be applied at this and all greater distances. This minimum test distance shall be determined by

taking illuminance readings over a series of increasing distances and then computing intensities using the ISL. Beyond a certain distance, the computed intensities will become constant. If it is more appropriate for the luminaire and its typical application, measurement can be made at a lesser distance; however, the limitations on the ISL shall be stated. *Note:* The minimum distance where the ISL applies to entertainment luminaires is often much greater than “five times effective aperture,” the common minimum distance suggested for general lighting luminaires.

2.6 Electrical Requirements

2.6.1 Lamp Regulation. For high intensity discharge (HID) lamps, lamp power shall be maintained at $\pm 0.5\%$ during the test and during lamp calibration for. Alternatively, luminaire voltage shall be maintained at $\pm 0.1\%$ for discharge lamp equipment.

Lamp current shall be maintained at $\pm 0.1\%$ for incandescent filament lamps.

2.6.2 Wave Form. The AC power supply shall have a sinusoidal voltage waveform such that the root-mean-square (RMS) summation of the harmonic components will not exceed 3.0% of the fundamental.

2.7 Electrical Instrumentation

2.7.1 Uncertainties. The measurement uncertainties of voltage and current shall be $\pm 0.5\%$ or better. The measurement uncertainties of wattage shall be $\pm 0.75\%$ or better. The actual uncertainties of electrical measurements shall be stated in the test report.⁵

Harmonics are present in discharge lamp circuits, and therefore instruments must respond adequately to frequencies considerably higher than the input line fundamental. The instrument(s) chosen for voltage and current measurements shall be of the true RMS type and shall have a specified accuracy and frequency response adequate to ensure that these requirements are met.⁶

Caution: When testing discharge lamps, digital test instruments and amplifiers are susceptible to burnout due to starting pulse voltages and high line voltages.

Therefore, necessary precautions must be taken so that such instruments are in the circuit only during lamp measurements and disconnected during lamp ignition.

2.7.2 Impedance Limitation. All measurements shall be corrected for the effects of measuring instrument impedance, if those effects are significant.

It is important to note that when the relative calibration method is used, meter impedance cancels between the luminaire and bare lamp tests.

2.8 Detectors

The most commonly used detector is the silicon cell. It is recommended because of its low noise, appropriate sensitivity, excellent stability and linearity, and fast response. Selenium detectors are now seldom used, as they are subject to fatigue, short life, temperature sensitivity, and non-linearity. The application of detectors and suitable measuring circuits is described in **Reference 6**. The detector must be provided with a filter that will bring its spectral response in close agreement with the CIE $V(\lambda)$ photopic spectral luminous efficiency curve⁷ (see **Annex A**). (See **Section 5.0** for information on the use of digital cameras for photometric testing.)

3.0 Preparation of Equipment for Test

3.1 Lamp Orientation

When a luminaire reflector and lamp are designed for a fixed orientation with respect to each other (as exists with a bi-post based lamp or a pre-focus based lamp), normal operating orientation shall be used during the photometric test. When the reflector-lamp relationship is not fixed (as exists with a screw-shell based lamp), the following test arrangements shall be adopted (unless data is required at alternative test positions, in which case the specific arrangement used shall be noted in the test report):

3.1.1 Filament Lamps. Filament lamp-reflector arrangements shall follow one of the configurations shown in **Figures 1a** through **1d**.⁸

**IES Approved Method for Photometric Testing of Entertainment Lighting Luminaires
Using Incandescent Filament Lamps or High Intensity Discharge Lamps**

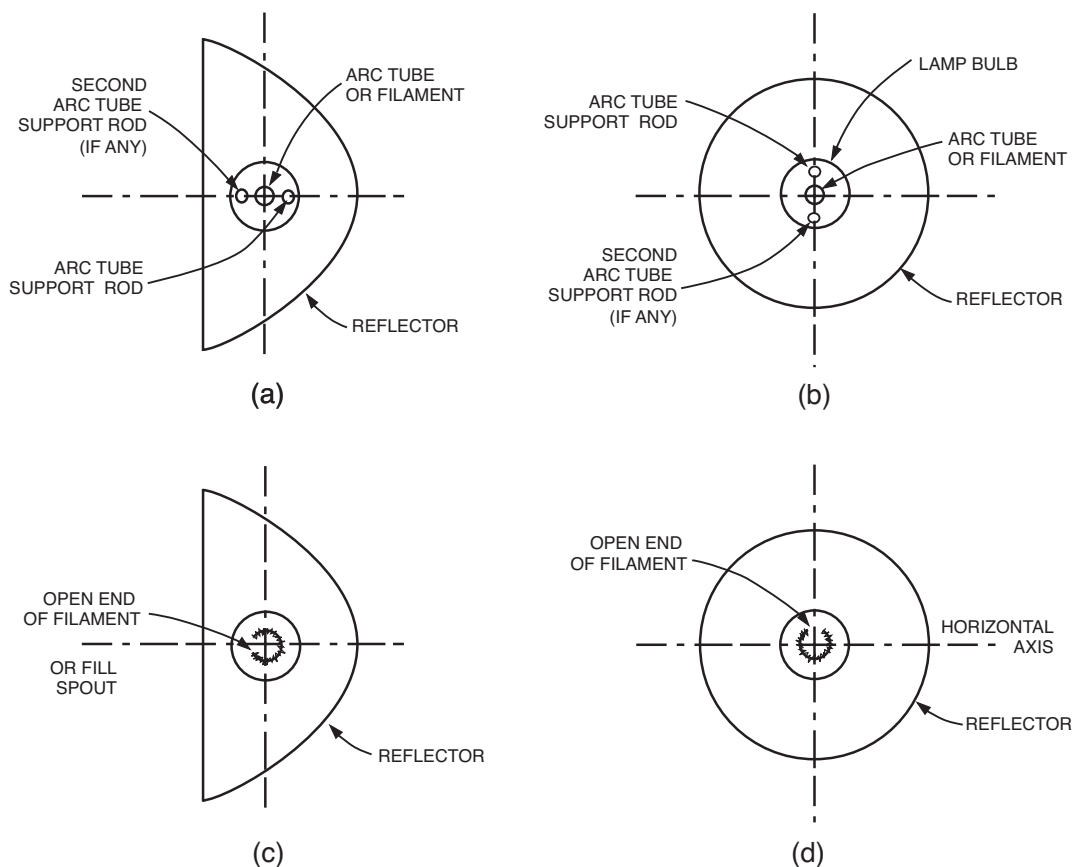


Figure 1. When a lamp-reflector relationship is not fixed, specific test arrangements shall be adopted during photometric tests. Illustrated in (a) and (b) are the correct lamp axis orientations for testing discharge lamps and linear filament lamps, while (c) and (d) describe the proper lamp axis orientations for testing lamps with curved, open-end filaments.

- For a lamp with its axis perpendicular to the reflector axis, the open end of the filament or the fill spout (tip-off) must point away from the principal reflector element (see **Figure 1c**).
- For a lamp with its axis parallel to the axis of the reflector, the open end of the filament must point toward the top of the reflector (see **Figure 1d**).
- When a filament lamp having a linear shaped filament (such as a CC8 filament) is used and the filaments axis is parallel to the principle reflector axis, the support rod must be toward the top of the reflector (see **Figure 1b**).
- For lamps with a linear shaped filament axis perpendicular to the principal axis of the reflector, the support rods shall lie in a plane parallel to the principle reflector axis. When only one support rod is present, it must lie towards the reflector (see **Figure 1a**).

3.1.2 Discharge Lamps. Discharge lamp-reflector arrangements shall follow the configuration shown in **Figure 1a** or **1b**.⁹

- When discharge lamps are used and the lamp axis is parallel to the principal reflector axis, tests shall be conducted with the arc tube support rod above the arc tube. If two arc tube supports are used, they must be on the vertical centerline (see **Figure 1b**).
- For discharge lamps having the lamp axis perpendicular to the principal axis of the reflector, tests shall be conducted with the arc tube support rod towards the reflector (see **Figure 1a**).

Metal halide lamps require special precautions. Long stabilization periods (up to several days) must be allowed when the operating orientation is changed. Even designated “universal operating” lamps require prolonged stabilization in their actual operating position

before they can be used as test lamps. In practice, separate lamps should be maintained for each lamp orientation (base up, base down, or horizontal) and should be operated only in their assigned orientation. Furthermore, the orientation of a horizontal operating lamp about its horizontal axis is critical. It is suggested that the lamp base be marked such that the mark will be positioned identically any time the lamp is energized. Metal halide lamps cannot be moved when they are turned off until all salts within the arc tube have completely cooled and solidified. For most applications, 30 minutes of cooling time is sufficient.

Single- and double-ended metal halide lamps without integral outer glass envelopes require special precautions and considerations. For example, some lamp types will not reach rated electrical, thermal, or photometric performance if operated in free air. A fixture simulator will usually be required for such lamps. Certain other lamps have quartz arc tubes and quartz outer envelopes that do not filter ultraviolet radiation. Because fixture simulators may not block this radiation, precautions must be taken to protect laboratory personnel from eye injury and actinic erythema (sunburn). There are also metal halide lamps that generate ozone, which can be toxic if inhaled in heavy concentrations. Adequate ventilation shall be provided. (See manufacturers' guidelines for specific information.)

3.2 Focusing

In fixed-focus units, the light center of the lamp must be placed at the design focal point of the reflector. The light center length of the lamp must be measured, and the lamp position adjusted if the test lamp does not have the same light center length as a nominal lamp.

In adjustable focus units, lamp position must be adjusted within the luminaire to produce the specific beam desired for test. The lamp position used for the test must be stated in the test report.

3.3 Cleaning of Optical Parts

All glass, reflectors, and optical parts shall be thoroughly cleaned before any measurements are made, unless the test is to determine how dirt buildup affects the luminaire. Any deviation shall be noted on the test report.

4.0 Testing by Goniophotometry

4.1 General

The goniophotometer provides a means of mounting the luminaire and measuring the detector angles relative to the luminaire. The goniophotometer must be rigid enough to provide correct angle measurements, even when there is appreciable off-balance loading. Goniophotometer construction shall provide for accurate angular settings that are reproducible to within 0.1 degrees.

4.2 Polar Axis Coordinates (Type B)

This Approved Method provides for type B data collection and reporting.¹⁰ (Type A goniophotometers may be used, but the angular positions used shall be the type A equivalents to the desired type B grid of angles. Data shall be reported as type B.)

Certain HID sources, such as metal halide, alter their output due to changes in lamp burning position. In this case, data shall be taken using a type C goniometer, which allows the burning orientation to be unchanged during the test (see **Section 4.4**). The data must then be converted to type B before the test report is prepared. This conversion must be noted in the report.

4.3 Positioning the Luminaire

The principal axis of a luminaire is defined as an imaginary line centered on the luminaire exit aperture (see **Figure 1 [Section 3.1.2]** and **Figure 2 [Section 4.4]**). The luminaire shall be aimed horizontally with its principal axis toward 0° horizontal. The "top" of the luminaire shall be oriented toward the zenith direction of the goniophotometer, and the "bottom" shall be oriented toward the nadir direction. The light center of the lamp should coincide with the goniophotometer center.

4.4 Moving Cell or Mirror Photometer (Type C)

The type C goniophotometer with a moving photodetector or mirror movable around a horizontal axis may be used.¹⁰ Angular directions are as shown in **Figure 2**. Tilting of the light source does not occur using type C equipment. Interpolation is necessary to produce data in a type B coordinate system.

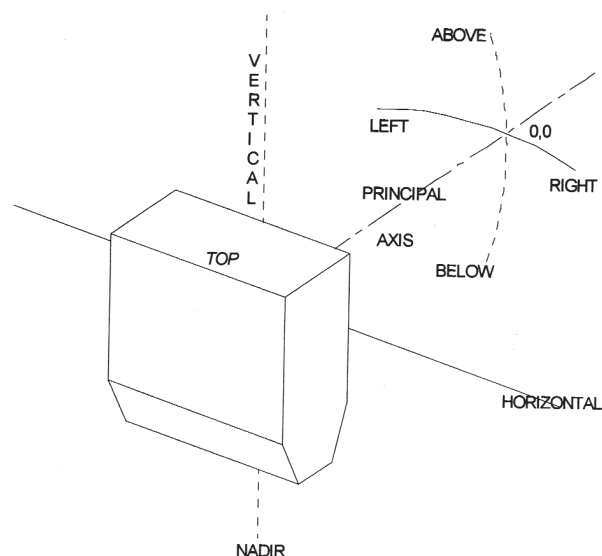


Figure 2. The defined angular directions for a luminaire have, as the principal axis, an imaginary line centered on (and perpendicular to) the reflector opening.

4.5 Calibration Procedure and Methods (General)

Either the *relative* method or the *absolute* method may be used for calibration.

The relative method shall be used when it is desired to state test results in terms of a lamp operating at initial rated light output. For test purposes, this method permits the use of any lamp (of the desired type) having the proper physical dimensions. The lamp need not operate at rated light output. The same instrumentation shall be used for measuring both the lamp and luminaire, minimizing the effects of test instrument response differences. Data shall be prorated to the manufacturer's rated lumens for that lamp type.

The absolute method shall be used when it is desired to determine the illumination characteristics of a specific luminaire together with a specific lamp under test, in terms of candelas and lumens directly traceable to national standards.

Lamp current or power shall be maintained by observing a calibrated meter. The selected current (for incandescent) or power (for HID) shall be held constant throughout the luminaire test and the lamp calibration, within the tolerances specified in **Section**

2.6.1. Alternatively, luminaire input voltage shall be held constant for tests involving HID sources.

4.5.1 Relative Calibration Method for Lamp and Goniophotometer. Certain HID sources require a controlled environment for thermal reasons, physical containment, and control of hazardous ultraviolet (UV) radiation. In such cases, relative photometry is not recommended. (See **Section 4.5.2** for methods of absolute calibration.)

In the relative calibration method, the relative total light output of the lamp under test shall be determined by summing the products of the relative luminous intensities and the appropriate solid angles (zonal constants). Readings are taken at vertical intervals of 5° or smaller, such as at 2.5, 7.5, 12.5, 17.5 degrees (up to 177.5 degrees), in ten or more equally spaced vertical half-planes around the lamp. Relative luminous intensities shall be measured by a system with linear response, usually not calibrated directly in candelas.

A calibration constant shall be established by taking the ratio of the *rated lumen output for the specific lamp type* to the *measured relative total lumen output of the bare test lamp*. Measured relative luminous intensities of the luminaire shall be multiplied by the constant to calculate the luminous intensities (in candelas) based on the rated lumens for the particular lamp used. Alternatively, an average of various manufacturers' listings of rated lumens shall be used.

Lamp orientation during calibration must be identical to that used during the luminaire test.

4.5.2 Absolute Calibration Method. The absolute calibration method is suitable for establishing luminaire performance on an absolute basis. This method shall be used where the lamp must operate in a controlled environment within the luminaire, and where bare lamp calibration is not possible. The method is also applicable to certain incandescent luminaires having performance dependent upon filament luminance rather than lamp lumens.

With absolute photometry, standard lamps of known luminous intensity in a specified direction shall be set

up on the photometer. A conversion constant must be established between the photometer's instrumentation readout and absolute candelas. At least three standard lamps shall be compared on the test photometer, with an average value used to establish the constant. The bare test lamp shall be checked against a luminous flux standard to determine its lumen output.¹¹

National Institute of Standards and Technology (NIST)-based absolute calibrated detectors (rather than standard lamps) can also be used.¹¹ For such a method, the photometer shall be calibrated by placing its photodetector on an optical bench adjacent to and on the same illuminated plane as an NIST-based photodetector.

To convert absolute lux (footcandle) values to absolute intensity in candelas, the lux value is multiplied by D^2/ρ , where D is the test distance measured using the same system of units the detector uses (SI or USCS), and ρ is the reflectance of the mirror (if one is used.)

This calibrated detector method has been stated to have twice the accuracy of the calibrated source method. The NIST-based photodetector must have been calibrated up to 100,000 lux (9,300 footcandles), or to whatever portion of this level will be encountered during measurement of the greatest intensity within the luminaire beam.

4.6 Selection of Angles for Photometric Measurements

Data angles shall be in 1-degree steps, or less, both horizontally and vertically, in type B coordinates, throughout the range of interest. In order to accurately compute total luminaire lumens, the angular range should be sufficient to include all data points that will generate a significant illuminance on a reference plane perpendicular to the optical axis of the luminaire. In practice, it is sufficient to collect data over an angular range where the intensity exceeds 3% of the maximum intensity.

When data is collected in type C coordinates, the vertical step size within each measurement plane shall be 1 degree or less, and the lateral step size between planes shall be 10 degrees or less.

4.7 Detector Size

To prevent integration of readings over a wide angular range, the photodetector shall subtend an angle no greater than 0.1° to the goniometric center.

5.0 Test Requirements for Specific Types of Luminaires

5.1 Fresnel-Lensed Types, Plano-Convex Types, Focusing Scoops, Ellipsoidal Spotlights, and Variable Ellipsoidal Follow Spots

All the named lensed and focusing-type luminaires must be tested for evenness of illuminated spot, which can be reported as a ratio of the maximum to minimum values. Such measurements shall be performed over the angular range of interest in angular step sizes not exceeding 1° . This analysis shall be conducted with adjustments set to produce maximum spot size, minimum spot size, and 50 percent spot size. *Note:* The Fresnel-type spots may have adjustable lamp positions or adjustable lens positions.

5.2 Ellipsoidal Reflector Luminaires

Many ellipsoidal reflector luminaires have adjustable socket positions allowing a beam uniformity adjustment range — from flat field to peaked center. Luminaires may be tested at different settings, and the setting used for each test shall be specified.

5.3 Asymmetric Luminaires

The reader is referred to LM-35, *IES Approved Method for Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps*,¹ or LM-41, *IES Approved Method for Photometric Testing of Indoor Fluorescent Luminaires*.¹²

5.4 Soft Light Floods

The reader is referred to LM-46, *IES Approved Method for Testing Indoor Luminaires Using High Intensity Discharge or Incandescent Filament Lamps*,⁴ or LM-41-14, *IES Approved Method for Photometric Testing of Indoor Fluorescent Luminaires*.¹²

5.5 Multiple-Lamp Luminaires

For a luminaire using multiple lamps, testing shall be performed with all lamps inside the luminaire and operating.

6.0 Computation of Test Results and Test Report

6.1 Photometric Data Processing

The following eight steps shall be used in developing information describing luminaire characteristics:

Step 1: When distribution is intended to be bilaterally symmetrical relative to the right and left sides of the luminaire, the corresponding intensity (candelas) in the right- and left-hand sides of the distribution shall be averaged.

Step 2: Create iso-illuminance (iso-lux or iso-footcandle) plot data from the measured values of intensity (candelas) for a surface perpendicular to the axis at a chosen distance.² Interpolation may be used as needed. The value of maximum illuminance shall be determined. Plot data shall then be computed for the following percentages of maximum illuminance: 3, 10, 20, 30, 40, 50, 60, 70, 80, and 90 percent (see **Annex B**).

Step 3: Compute the lumens in each zone using zonal constants (see **Annex C**). Compute the *total lumen output* by summing the lumens in all zones falling within the area of the illuminance diagram that are illuminated to a level of 3% or more of the peak illuminance in the same plane. (Calculate the illuminance at each respective zone center on the reference plane. If the value exceeds 3% of the maximum illuminance, include the zonal lumens for that zone in the sum). *Total lumen output* is identical to *cutoff lumen output*, used in entertainment lighting.

Step 4: Compute *total efficiency* by dividing the total lumen output of the luminaire by the lamp lumens.

Step 5: Obtain the *half-peak lumens* by summing the lumens in all zones that have a center point lying within the 50%-of-maximum illuminance contour (see **Annex B**). (Calculate the illuminance at each respective zone center on the reference plane. If the value exceeds 50% of the maximum illuminance, include the zonal lumens for that zone in the sum.)

Step 6: Obtain the *tenth-peak lumens* by summing the lumens in all zones that have a center point lying within the 10%-of-maximum illuminance contour (see **Annex B**). (Calculate the illuminance at each respective zone center on the reference plane. If the value exceeds 10% of the maximum illuminance, include the zonal lumens for that zone in the sum.)

Step 7: For luminaires with a conical beam distribution, determine the half-peak illuminance angle, tenth-peak illuminance angle, and cutoff illuminance angle by computing the total subtended angle between locations of 50%, 10%, and 3% of maximum illuminance, respectively, on the left and right sides of the beam.

Step 8: Prepare plotted data for a rectilinear plot of intensity versus angle for a vertical and horizontal plane through the beam axis. The angular step sizes shall be selected in accordance with the method given in **Section 4.6**.

6.2 Test Report

The test report shall conform to the requirements of the Entertainment Services and Technology Association (ESTA) document ANSI E1.9-2007/R2017, *Reporting Photometric Performance Data for Luminaires Used in Entertainment Lighting*.²

The report shall note any deviations from ANSI/IES LM-73-18(R2023).

Data may be written electronically in a file conforming to the IES standard format.¹³

Annex A — Measurement Errors Due to the Deviation In System Response from $V(\lambda)$

NOTE: This annex is not part of ANSI/IES LM-73-18(R2023). It is included for informational purposes only. References are found at the end of this annex.

Commercially available photometric detectors only approximate the standard $V(\lambda)$ response, due to limitations in the manufacturing process. Generally, the closer the “fit” to $V(\lambda)$, the more expensive the detector. A numerical term used to express the accuracy of fit is f_1' .^{a,b} This term provides an indication of the *average* closeness of the detector response to the ideal curve (i.e., its overall conformance to the standard). However, f_1' does not provide information about the magnitude of response deviation from $V(\lambda)$ at specific wavelengths. To determine f_1' , the deviation from the $V(\lambda)$ curve is determined at a series of wavelengths, and the absolute values of these deviations are integrated across the visible range.

To determine the error of a particular measurement due to detector spectral mismatch, the following must be known:

- The relative spectral response of the detector
- The spectral distribution of the source being measured
- The spectral distribution of the source used to calibrate the detector

The detector’s relative spectral response is often available from the manufacturer. It can also be measured in the U.S. at the National Institute of Standards and Technology, or at any of several commercial laboratories, provide the necessary equipment and the expertise are available.^c The spectral distributions of the source under test and the calibration source must be determined using a spectroradiometer. A correction can then be calculated and applied.^d

REFERENCES FOR ANNEX A

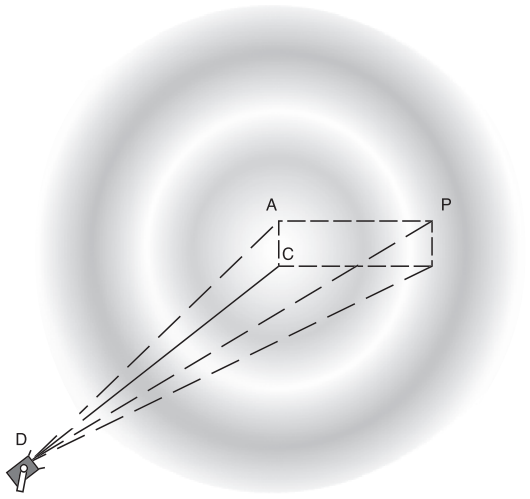
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Annex B – Calculations of Illuminance Values for Type B Coordinate System

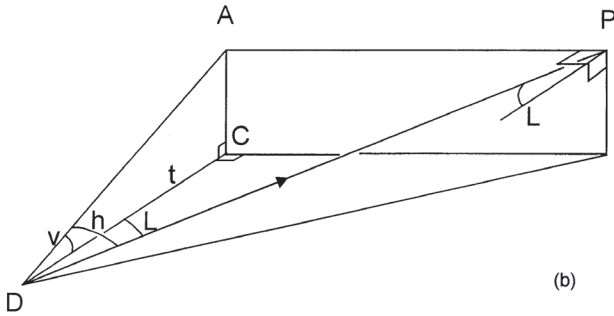
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Referring to **Figure B-1**:

- The luminaire is at point *D* and is aimed towards point *C*, which lies on a flat surface perpendicular to luminaire axis *DC*. The illuminance calculation point is *P*.
- The origin of the light pattern is at *C* (0, 0 location).
- The photometric vertical angle for point *P* is defined as *v* (type B coordinate).
- The photometric horizontal angle for point *P* is defined as *h* (type B coordinate).
- The throw distance is defined as *t* and is equal to distance *CD*.



(a)



(b)

Figure B-1. In (a), a spotlight is assumed to be aimed at a flat wall. It is desired to find illuminance at point *P* using a type B coordinate system. The key points, distances, and angular relationships within the light beam are detailed in (b).

In addition:

- The vertical throw multiplier for point *P* is *a*, defined as:

$$a = \frac{AC}{CD}$$

- The horizontal throw multiplier for point *P* is *b*, defined as:

$$b = \frac{AP}{CD}$$

To utilize the inverse-square law, three quantities must be known: the intensity at angles *h* and *v* (type B coordinates), the throw distance *t*, and the angle of incidence, *L*. These are used to find the illuminance at point *P*. (Refer to the ANSI/IES LS-4-20, for a complete discussion of the inverse-square law).

Step 1. Determine the intensity I_{hv} at point *P* by finding angle *v* in terms of *a*, and angle *h* in terms of *b*, and then consulting the type B coordinate intensity data array:

Find angle *v* from the geometry of **Figure B-1**:

$$v = \text{atan}\left(\frac{AC}{CD}\right)$$

Substituting from the definition given for the vertical throw multiplier *a*, the equation for angle *v* becomes:

$$v = \text{atan}(a) \quad (\text{B-1})$$

Find angle *h* from the geometry of **Figure B-1**:

$$h = \text{atan}\left(\frac{AP}{AD}\right)$$

However, it is important to note that:

$$\cos v = \left(\frac{CD}{AD}\right)$$

Rearranging terms and solving for *AD* yields:

$$AD = \frac{CD}{\cos v}$$

Replacing *AD* in the initial equation for angle *h* with the above expression yields:

$$h = \text{atan}\left[\frac{AP}{CD}(\cos v)\right]$$

Substituting from the definition given for the horizontal throw multiplier *b*, the equation for angle *h* becomes:

$$h = \text{atan}(a \cos v) \quad (\text{B-2})$$

Angle *v* and angle *h* can now be found from **Equation B-1** and **Equation B-2** respectively, and the intensity to point *P* (I_{hv}) is determined from the type B coordinate intensity data array.

Step 2. Compute angle of incidence, *L*, which may be conveniently found from the determination of $\cos L$. From the geometry of **Figure B-1** it is determined that:

$$\cos L = \frac{CD}{PD}$$

PD can be shown equal to:

$$PD = \sqrt{(AC)^2 + (AP)^2 + (CD)^2}$$

After substituting from the initial definitions given for the vertical throw multiplier a and the horizontal throw multiplier b , the equation for PD becomes:

$$PD = \sqrt{(aCD)^2 + (bCD)^2 + (CD)^2}$$

This expression for PD is substituted into the initial equation for $\cos L$, yielding:

$$\cos L = \frac{CD}{\sqrt{(aCD)^2 + (bCD)^2 + (CD)^2}}$$

or:

$$\cos L = \frac{CD}{\sqrt{CD^2 a^2 + b^2 + 1}}$$

Thus:

$$\cos L = \frac{1}{\sqrt{a^2 + b^2 + 1}} \quad (\text{B-3})$$

and:

$$L = \cos^{-1} \left[\frac{1}{\sqrt{a^2 + b^2 + 1}} \right]$$

Step 3. Compute illuminance E on the illuminated plane at point P , where the intensity is I_{hv} .

The inverse-square law says that the illuminance E at a point on a surface at distance d from a source having luminous intensity I is found from:

$$E = \frac{I}{d^2}$$

When the point on a surface is not normal to the incident light, the angle of incidence θ is accounted for by the cosine law, which may be combined with the inverse-square law as follows:

$$E = \frac{I}{d^2} \cos \theta \quad (\text{B-4})$$

Referring to the geometry of **Figure B-1**, it is desired to find the illuminance E for point P . Relative to the luminaire (source) at D , this point is located at distance PD and is offset from the normal by angle L . The luminaire has a luminous intensity of I_{hv} . Substituting these values into **Equation B-4**:

$$E = \frac{I_{hv} \cos L}{(PD)^2} \quad (\text{B-5})$$

However, from **Figure B-1**, it can be seen that $PD = CD / \cos L$. Substituting this expression for PD into **Equation B-5**:

$$E = \frac{I_{hv} \cos L}{(CD / \cos L)^2}$$

This can be rewritten as:

$$E = \frac{I_{hv} \cos^3 L}{(CD)^2} \quad (\text{B-6})$$

(The above is commonly known as the cosine-cubed law.)

Taking the expression that was developed for $\cos L$ in **Equation B-3** and cubing it yields:

$$\cos^3 L = \frac{1}{(a^2 + b^2 + 1)^{3/2}}$$

From substitution, and noting from the initial definitions that CD is equal to the throw distance t , **Equation B-6** becomes:

$$E = \frac{I_{hv}}{t^2 (a^2 + b^2 + 1)^{3/2}} \quad (\text{B-7})$$

Note: Where illuminance values are to be expressed as a percentage of the maximum, the value of t may be set to 1.

Annex C — Calculation of Lumens

NOTE: This annex is not part of ANSI/IES LM-73-18(R2023). It is included for informational purposes only.

Luminaire efficiency in percent is obtained by dividing the total luminaire lumens by the lamp lumens and multiplying the result by 100. Each measured intensity value is assumed to lie at the center of a zone, with each zone having a size defined by its horizontal and vertical angular widths, these being determined in accordance with **Section 4.6** of this document. Within a given zone, the intensity is assumed constant. Zonal lumens are determined by multiplying the center zone intensity by the zonal constant (solid angle) for that zone. Zonal constants for a type B coordinate system, are found using:

$$Z = \frac{V\pi(\sin H1 - \sin H2)}{180}$$

where:

Z = zonal constant

V = the vertical increment (in degrees)

$H1$ and $H2$ = the horizontal angles of the zone boundaries (in degrees)

Summing the products of center zone intensity and Z (for all zones) completes the calculation of total lumens.

For the computation of half-peak lumens, the illuminance projected on a plane perpendicular to the luminaire axis is computed at points corresponding to the horizontal and vertical angles of the center of each zone (see **Annex B**). Any value may be chosen for the distance from the luminaire to the plane. The maximum illuminance on this plane also is computed. If the illuminance on the plane at the center of a given zone is equal to or greater than the maximum illuminance, the lumens of that zone are included in the count of half-peak lumens.

Similarly, the one-tenth-peak lumens and cutoff lumens are determined based on 10% and 3%, respectively, of the maximum illuminance.

Annex D – Glossary

*Note: The definitions given in this glossary for cutoff lumen output, half-peak lumens, one-tenth-peak illuminance angle, and one tenth peak lumens are in accordance with accepted practice in the entertainment industry, rather than lumen levels as presented in LM-35, IES Approved Method for Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps (see **Reference 1** in the main document).*

bit: A single digit in a binary number system, either 0 or 1.

CCD camera: A type of camera that utilizes a silicon-based light detector termed a charge-coupled device

(CCD), which consists of an array of individual detector elements.

CIE $V(\lambda)$ curve: The sensitivity response curve of the human eye to light of differing wavelengths.

cutoff field area: The area on a plane normal to the optical axis of the luminaire and illuminated by the luminaire, in which the level of illuminance is 3% or more of the maximum illuminance in the same plane. Throw distance must be specified.

cutoff illuminance angle: The angle subtended to the luminaire by points on an illuminated plane normal to the optical axis of the luminaire, where the illuminance is 3% of the maximum illuminance in the same plane. The 3% points are defined as points on the illuminated plane on either end of a horizontal line which passes through the point on the illuminated plane intercepted by the optical axis of the luminaire (see **Figure D-1**).

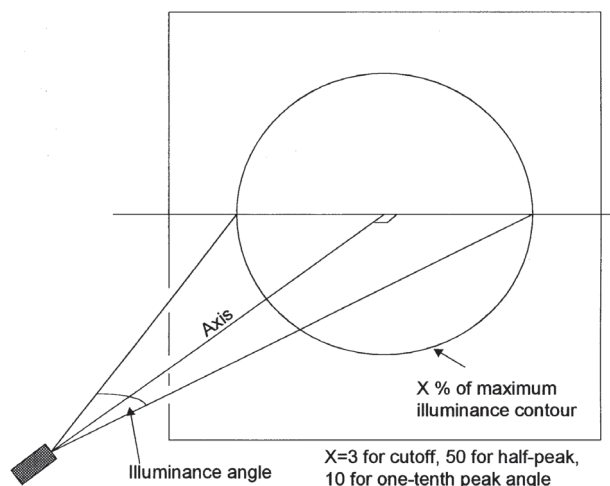


Figure D-1. This diagram describes the angle subtended to the luminaire by points on an illuminated plane. Point locations form a spherical pattern determined by the specific “percent of maximum illuminance contour” (value of X) under consideration.

cutoff lumen output: The lumens falling within the area of an iso-illuminance diagram that is illuminated to a level of 3 percent or more of the peak illuminance in the same plane. (See also *total lumen output*).

digitization: The process by which analog data are converted into numerical data.

exit aperture: The luminous opening of a luminaire through which the light beam is emitted. With multiple lamp or multiple compartment luminaires, the exit aperture shall be taken as the entire array of lamps or compartments.

far-field luminaire: A luminaire whose intended operating distance from the subject is greater than or equal to the distance at which the inverse-square law can be used to predict an illuminance level. (For narrow-beam equipment, this may be considerably more than the normally used distance of five times the maximum luminous dimension of the luminaire.)

half-peak illuminance angle: The angle subtended to the luminaire by points on an illuminated plane normal to the optical axis of the luminaire, where the illuminance is 50% of the maximum illuminance in the same plane. The 50% points are defined as points on the illuminated plane on either end of a horizontal line which passes through the point on the illuminated plane intercepted by the optical axis of the luminaire (see **Figure D-1**).

half-peak lumens: The lumens falling within the area of an iso-illuminance diagram in which the level of illuminance is 50 percent or more of the maximum illuminance in the same plane.

hard-focus: A focus position that achieves the most clearly defined edge to the illuminated area.

illuminance: The areal density of the luminous flux incident at a point on a surface.

iso-illuminance diagram: A series of iso-illuminance lines for various illuminance values plotted on a common graph (see **Figure D-2**).

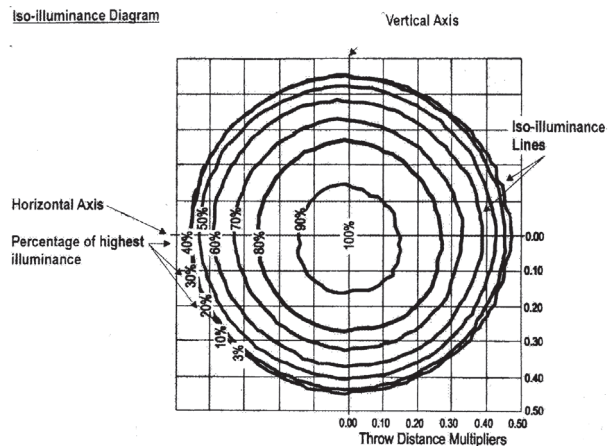


Figure D-2. Like the non-intersecting lines on a weather map connecting all places with the same barometric pressure, this iso-illuminance diagram connects together all points on a surface having equal illuminance (e.g., the approximately circular “70%” line connects all points receiving 70% of the highest illuminance value).

iso-illuminance line: A line plotted on a set of coordinates to show all the points on an illuminated surface where illuminance is the same.

light beam: The light emitted from the exit aperture of a luminaire.

luminaire efficacy: The ratio of a luminaire’s total lumen output divided by the power consumed, expressed in terms of lumens per watt.

luminaire efficiency: The ratio of a luminaire’s total lumen output to the lumens emitted by the lamp, expressed as a percentage.

luminaire: A complete lighting unit, consisting of a lamp or lamps, together with all the parts that are needed to position and protect the lamp or lamps, distribute the light, and connect the lamp or lamps to the input power. For HID luminaires, this may include an integral ballast.

near-field luminaire: A luminaire whose intended operating distance from the subject is shorter than the minimum distance at which the inverse-square law can be used to predict an illuminance level.

one-tenth-peak illuminance angle: The angle subtended to the luminaire by points on an illuminated plane normal to the optical axis of the luminaire, where the illuminance is 10% of the maximum illuminance in the same plane. The 10% points are defined as points on the illuminated plane on either end of a horizontal line which passes through the point on the illuminated plane intercepted by the optical axis of the luminaire (see **Figure D-1**).

one-tenth-peak lumens: The lumens falling within the area of an iso-illuminance diagram that is illuminated to a level of 10% or more of the maximum illuminance in the same plane.

throw distance: The distance between the exit aperture of the luminaire and the surface that is being illuminated.

throw distance multiplier: A number that when multiplied by the throw distance yields the distance of a point in the cutoff field area from the origin.

total lumen output: In the entertainment industry: the lumens falling within the area of an iso-illuminance diagram that is illuminated to a level of 3% or more of the peak illuminance in the same plane. This is identical to *cutoff lumen output*. *Note:* This does not correspond to standard IES usage of the term, which describes the total light output at all angles.

variable angle luminaire: A luminaire that has optical elements designed to be adjusted to vary the total area illuminated at a given throw distance.

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